22/01/2025
Evening

Aakash
Medical | IIT-JEE | Foundations

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Answers & Solutions

Time : 3 hrs. *for* M.M. : 300

JEE (Main)-2025 Phase-1

[Computer Based Test (CBT) mode]

(Mathematics, Physics and Chemistry)

IMPORTANT INSTRUCTIONS:

(1) The test is of **3 hours** duration.

(2) This test paper consists of 75 questions. Each subject (MPC) has 25 questions. The maximum marks are 300.

(3) This question paper contains **Three** Parts. **Part-A** is Physics, **Part-B** is Chemistry and **Part-C** is **Mathematics**. Each part has only two sections: **Section-A** and **Section-B**.

(4) **Section - A :** Attempt all questions.

(5) **Section - B :**Attempt all questions.

(6) **Section - A (01 – 20)** contains 20 multiple choice questions which have **only one correct answer**. Each question carries **+4 marks** for correct answer and **–1 mark** for wrong answer.

(7) **Section - B (21 – 25) contains** 5 **Numerical value** based questions. The answer to each question should be rounded off to the **nearest integer**. Each question carries **+4 marks** for correct answer and **–1 mark** for wrong answer.



MATHEMATICS

SECTION - A

Multiple Choice Questions: This section contains 20 multiple choice questions. Each question has 4 choices (1), (2), (3) and (4), out of which **ONLY ONE** is correct.

Choose the correct answer:

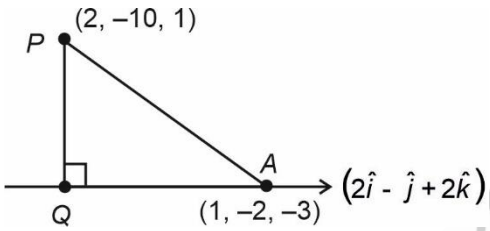
1. The perpendicular distance, of the line $\frac{x-1}{2}=\frac{y+2}{-1}=\frac{z+3}{2}$ from the point $P(2,-10,1)$, is:

(1) $4\sqrt{3}$ (2) $3\sqrt{5}$

(3) $5\sqrt{2}$ (4) 6

Answer (2)

Sol.



PQ (shortest distance)

QA will be projection of PA on line

$\overrightarrow{PA}=(-\hat{i}+8\hat{j}-4\hat{k}) \Rightarrow |\overrightarrow{PA}|=\sqrt{81}=9$

$$\text{Projection}=\frac{|-2-8-8|}{\sqrt{2^2+(-1)^2+2^2}}=\frac{18}{3}=6$$

$PQ^2 = PA^2 - QA^2 = 81 - 36 = 45$

$\Rightarrow PQ=\sqrt{45}=3\sqrt{5}$

2. Let $\vec{a}$ and $\vec{b}$ be two unit vectors such that the angle between them is $\frac{\pi}{3}$. If $\lambda\vec{a}+2\vec{b}$ and $3\vec{a}-\lambda\vec{b}$ are perpendicular to each other, then the number of values of λ in $[-1, 3]$ is:

(1) 1 (2) 2

(3) 3 (4) 0

Answer (4)

Sol. $(\lambda\vec{a}+2\vec{b})\cdot(3\vec{a}-\lambda\vec{b})=0$

$\Rightarrow (3\lambda-2\lambda)+\vec{a}\cdot\vec{b}(-\lambda^2+6)=0$

since $\vec{a}\cdot\vec{b}=(1)(1)\cos\left(\frac{\pi}{3}\right)=\frac{1}{2}$

$\Rightarrow (-\lambda^2+6)\frac{1}{2}+\lambda=0 \Rightarrow \lambda^2-2\lambda-6=0$

$(\lambda-1)^2=7$

$\lambda=\pm\sqrt{7}+1\notin[-1,3]$

$\Rightarrow$ no values

3. If $\lim\limits_{x\to\infty}\left(\left(\frac{e}{1-e}\right)\left(\frac{1}{e}-\frac{x}{1+x}\right)\right)^x=\alpha$, then the value of $\frac{\log_e\alpha}{1+\log_e\alpha}$ equals:

(1) e^{-2} (2) e

(3) e^{-1} (4) e^2

Answer (2)

Sol. $\lim\limits_{x\to\infty}\left(\left(\frac{e}{1-e}\right)\left(\frac{1}{e}-\frac{x}{1+x}\right)\right)^x=(1^\infty \text{ from})$

$$\lim_{x\to\infty}\left[1+\left(\frac{e}{1-e}\right)\left[\frac{1}{e}-\frac{x}{1+x}\right]-1\right]^x$$

$\Rightarrow \alpha=e^{\lim\limits_{x\to\infty}\left[\left(\frac{e}{1-e}\right)\left[\frac{1}{e}-\frac{x}{1+x}\right]-1\right]x}$

$\Rightarrow e^{\lim\limits_{x\to\infty}\left[\left(\frac{1}{1-e}-1\right)+\frac{ex}{(e-1)(1+x)}\right]x}$

$\Rightarrow e^{\lim\limits_{x\to\infty}\left[\frac{e}{1-e}+\frac{ex}{(e-1)(1+x)}\right]x}$

$\Rightarrow e^{\lim\limits_{x\to\infty}\left(\frac{e}{e-1}\right)\left[\frac{x}{1+x}-1\right]x}=e^{\lim\limits_{x\to\infty}\frac{-ex}{(e-1)(1+x)}}=e^{\frac{-e}{e-1}}$



$$\Rightarrow \alpha = e^{\frac{e}{1-e}} \Rightarrow \ln\alpha = \frac{e}{1-e}$$

$$\Rightarrow \frac{\ln\alpha}{1+\ln\alpha} = \frac{\frac{e}{1-e}}{1+\frac{e}{1-e}} = e$$

4. Let α, β, γ and δ be the coefficients of x^7, x^5, x^3 and x respectively in the expansion of $\left(x+\sqrt{x^3-1}\right)^5 + \left(x-\sqrt{x^3-1}\right)^5$, $x > 1$. If u and v satisfy the equations

$\alpha u + \beta v = 18$,

$\gamma u + \delta v = 20$,

then $u + v$ equals:

(1) 3

(2) 4

(3) 5

(4) 8

Answer (3)

Sol. $\left(x+\sqrt{x^3-1}\right)^5 + \left(x-\sqrt{x^3-1}\right)^5$

$$= \left[{}^5C_0x^5 + {}^5C_1x^4\left(\sqrt{x^3-1}\right) + \ldots\ldots + {}^5C_5\left(\sqrt{x^3-1}\right)^5\right] +$$

$$\left[{}^5C_0x^5 - {}^5C_1x^4\left(\sqrt{x^3-1}\right) + \ldots\ldots + {}^{-5}C_5\left(\sqrt{x^3-1}\right)^5\right]$$

$$= 2\left[x^5 + {}^5C_2x^3\left(x^3-1\right) + {}^5C_4\left(x^3-1\right)^2\right]$$

$$= 10x^7 + 20x^6 + 2x^5 - 20x^4 - 20x^3 + 10x$$

Now

$\alpha = 10, \beta = 2, \gamma = -20, \delta = 10$

Also,

$$\left.\begin{aligned} 10u + 2v &= 18 \\ -20u + 10v &= 20 \end{aligned}\right\} u = 1,\ v = 4$$

$u + v = 5$

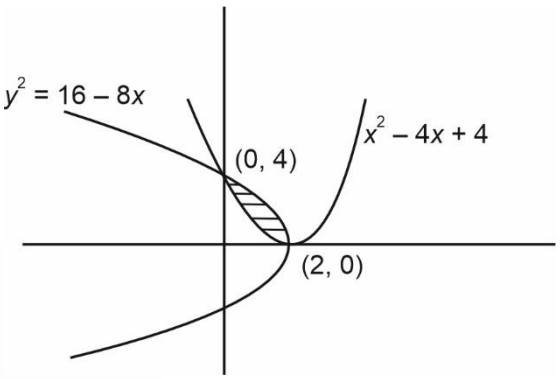
5. The area of the region enclosed by the curves $y = x^2 - 4x + 4$ and $y^2 = 16 - 8x$ is

(1) $\frac{8}{3}$ (2) 5

(3) $\frac{4}{3}$ (4) 8

Answer (1)

Sol.



$$\text{Area} = \int_0^2 \left(\sqrt{16-8x} - \left(x^2 - 4x + 4\right)\right)dx$$

$$= \frac{-(16-8x)^{3/2}}{12} - \frac{x^3}{3} + 2x^2 + 4x\Bigg]_0^2$$

$$= \frac{8}{3}$$

6. Let $A = \{1, 2, 3, 4\}$ and $B = \{1, 4, 9, 16\}$. Then the number of many-one functions $f: A \to B$ such that $1 \in f(A)$ is equal to

(1) 163 (2) 127

(3) 151 (4) 139

Answer (3)

Sol. $A = \{1, 2, 3, 4\}$

$B = \{1, 4, 9, 16\}$

Total number of functions = 4^4

Total number of one-one functions = 4!

Total number of many one functions = $4^4 - 4! = 232$



Total number of many-one functions in which $1 \notin f(A)$ $= 3 \times 3 \times 3 \times 3 = 81$

$\therefore$ Total number of many one functions $1 \notin f(A)$

$= 232 - 81$

$= 151$

7. For a 3×3 matrix M, let trace (M) denote the sum of all the diagonal elements of M. Let A be a 3×3 matrix such that $|A| = \frac{1}{2}$ and trace (A) = 3. If B = adj(adj($2A$)), then the value of $|B|$ + trace (B) equals

(1) 174 (2) 132

(3) 56 (4) 280

Answer (4)

Sol. $\because$ $\text{tr}(A) = 3$ and $|A| = \frac{1}{2}$

Now, $B = \text{adj}(\text{adj}(2A)) = |2A|^{3-2} \cdot (2A)$

$= 2^3 |A| \cdot 2A = 8A$

$\therefore$ $\text{tr}(B) = 8\ \text{tr}(A) = 24$

and $|B| = |8A| = 8^3 \cdot \frac{1}{2} = 256$

$\therefore$ trace (B) + $|B|$ = 24 + 256

= 280

8. Let a line pass through two distinct points $P(-2, -1, 3)$ and Q, and be parallel to the vector $3\hat{i} + 2\hat{j} + 2\hat{k}$. If the distance of the point Q from the point $R(1, 3, 3)$ is 5, then the square of the area of ΔPQR is equal to

(1) 144 (2) 136

(3) 140 (4) 148

Answer (2)

Sol. Equation of line PQ is:

$$\frac{x+2}{3} = \frac{y+1}{2} = \frac{z-3}{2} = r(\text{say})$$

Let coordinate of $Q = (3r - 2, 2r - 1, 2r + 3)$

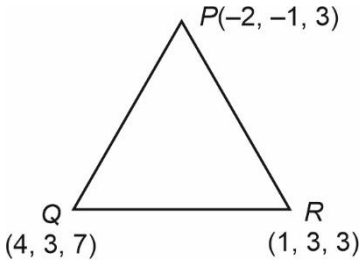
$\because$ $PR = 5$

Then

$$(3r-2-1)^2 + (2r-1-3)^2 + (2r+3-3)^2 = 25$$

$\therefore$ $r = 0$ or 2

$\therefore$ Coordinate of $Q = (4, 3, 7)$



$\therefore$ square of area of $\Delta PQR = \left|\frac{1}{2}\left(\overrightarrow{PQ} \times \overrightarrow{PR}\right)\right|^2$

$= \left|\frac{1}{2}\left(6\hat{i} + 4\hat{j} + 4\hat{k}\right) \times \left(3\hat{i} + 4\hat{j}\right)\right|^2$

$= \left|-8\hat{i} + 6\hat{j} + 6\hat{k}\right|^2 = 136$

9. If the system of linear equations:

$x + y + 2z = 6$,

$2x + 3y + az = a + 1$,

$-x - 3y + bz = 2b$,

where $a, b \in \mathbf{R}$, has infinitely many solutions, then $7a + 3b$ is equal to:

(1) 9 (2) 22

(3) 16 (4) 12

Answer (3)

Sol. The given equations are

$x + y + 2z = 6$,

$2x + 3y + az = a + 1$

$-x - 3y + bz = 2b$, where a, b, $\in$ **R**.

For infinite many solutions:

$D = D_1 = D_2 = D_3 = 0$



$$\therefore\ D=\begin{vmatrix}1 & 1 & 2\\ 2 & 3 & a\\ -1 & -3 & b\end{vmatrix}=2a+b-6$$

$$D_1=\begin{vmatrix}6 & 1 & 2\\ a+1 & 3 & a\\ 2b & -3 & b\end{vmatrix}=12a+5b+ab-6$$

$$D_2=\begin{vmatrix}1 & 6 & 2\\ 2 & a+1 & a\\ -1 & 2b & b\end{vmatrix}=-4a-3b-ab+2$$

and $D_3=\begin{vmatrix}1 & 1 & 6\\ 2 & 3 & a+1\\ -1 & -3 & 2b\end{vmatrix}=2a+2b-16$

from above relations

$a=-2,\ b=10$

$\therefore\ \ 7a+3b=16$

10. Let the curve $z(1+i)+\bar{z}(1-i)=4,\ z\in C$, divide the region $|z-3|\le 1$ into two parts of areas α and β. Then $|\alpha-\beta|$ equals:

(1) $1+\frac{\pi}{4}$ (2) $1+\frac{\pi}{3}$

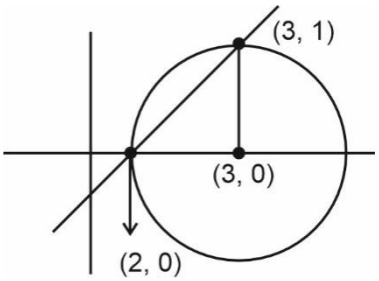
(3) $1+\frac{\pi}{6}$ (4) $1+\frac{\pi}{2}$

Answer (4)

Sol. Put $z=x+iy$

$(x+iy)(1+i)=(x-iy)(1-i)=4$

$\Rightarrow x-y=2$



Area of circle $=\pi$

Area of smaller region $=\frac{\pi}{4}-\frac{1}{2}$

Area of larger region $=\frac{3\pi}{4}+\frac{1}{2}$

$|\alpha-\beta|=1+\frac{\pi}{2}$

11. Let $P\left(4,4\sqrt{3}\right)$ be a point on the parabola $y^2=4ax$ and PQ be a focal chord of the parabola. If M and N are the foot of perpendiculars drawn from P and Q respectively on the directrix of the parabola, then the area of the quadrilateral $PQMN$ is equal to:

(1) $\frac{263\sqrt{3}}{8}$ (2) $\frac{343\sqrt{3}}{8}$

(3) $\frac{34\sqrt{3}}{3}$ (4) $17\sqrt{3}$

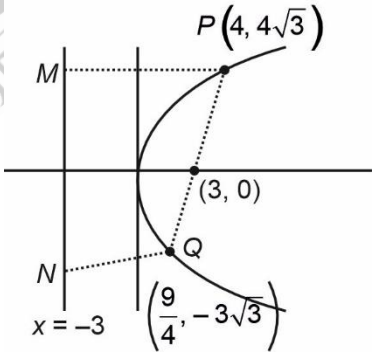
Answer (2)

Sol. $y^2=4ax$

$\left(4,4\sqrt{3}\right)$ lies on parabola.

$\Rightarrow a=3$

$y^2=12x$ $\qquad y=4\sqrt{3}x-12\sqrt{3}$



$ar(\square PQNM)=\frac{\left(7+\frac{21}{4}\right).7\sqrt{3}}{2}$

$=\frac{343\sqrt{3}}{8}$



12. In a group of 3 girls and 4 boys, there are two boys B_1 and B_2. The number of ways, in which these girls and boys can stand in a queue such that all the girls stand together, all the boys stand together, but B_1 and B_2 are not adjacent to each other, is :

(1) 120 (2) 96

(3) 144 (4) 72

Answer (3)

Sol. $\boxed{G_1}\boxed{G_2}\boxed{G_3}_\, B_3 _\, B_4 _$

$2 \times 2 \times 3! \times 2! \times {}^3C_2 = 144$

13. Let $f(x) = \int_0^{x^2} \frac{t^2 - 8t + 15}{e^t}\, dt,\ x \in R$. Then the numbers of local maximum and local minimum points of f, respectively, are:

(1) 2 and 3 (2) 1 and 3

(3) 2 and 2 (4) 3 and 2

Answer (1)

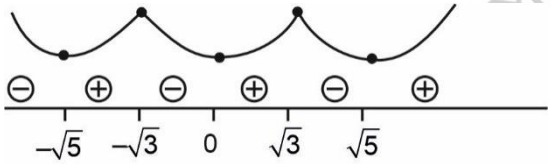
Sol. $f(x) = \int_0^{x^2} \frac{t^2 - 8t + 15}{e^t}\, dt,\ x \in R$

$$f'(x) = \frac{x^4 - 8x^2 + 15}{e^{x^2}}(2x) = 0$$

$$\Rightarrow \frac{2 \times (x^2 - 5)(x^2 - 3)}{e^{x^2}} = 0$$

$$\Rightarrow x(x + \sqrt{5})(x - \sqrt{5})(x + \sqrt{3})(x - \sqrt{3}) = 0$$

By using wavy curve method



Number of local maximum = 2

Number of local minimum = 3

$\Rightarrow$ option (1) is correct

14. Suppose that the number of terms in an A.P. is $2k$, $k \in N$. If the sum of all odd terms of the A.P. is 40, the sum of all even terms is 55 and the last term of the A.P. exceeds the first term by 27, then k is equal to :

(1) 8

(2) 4

(3) 5

(4) 6

Answer (3)

Sol. Let the A.P. be

$a, a + 2, a + 2d, \ldots, a + (2k - 1)d$

Now, $a + a + 2d + a + 4d + \ldots + a + (2k - 2)d = 40$

$Ka + 2d + 4d + \ldots + (2k - 2)d = 40$

$$\Rightarrow ka + \frac{k-1}{2}[2d + 2kd - 2d] = 40$$

$$\Rightarrow ka + k(k-1)d = 40 \qquad \ldots(1)$$

And $a + d + a + 3d + \ldots + a + (2k-1)d = 55$

$$\Rightarrow ka + \frac{k}{2}(d + 2kd - d) = 55$$

$$\Rightarrow ka + k^2 d = 55 \qquad \ldots(2)$$

Also, $a + (2k - 1)d - a = 27$

$$\Rightarrow (2k-1)d = 27 \Rightarrow d = \frac{27}{2k-1} \qquad \ldots(3)$$

From equation (1) and (2)

$k^2 d - kd - k^2 d = -15$

$$\Rightarrow d = \frac{15}{k} \qquad \ldots(4)$$

From equation (3) and (4)

$$\frac{27}{2k-1} = \frac{15}{k}$$

$27k = 30k - 15$

$\Rightarrow 3k = 15$

$\Rightarrow \boxed{k = 5}$



15. If A and B are two events such that $P(A \cap B) = 0.1$, and $P(A|B)$ and $P(B|A)$ are the roots of the equation $12x^2 - 7x + 1 = 0$, then the value of $\frac{P(\bar{A} \cup \bar{B})}{P(\bar{A} \cap \bar{B})}$ is :

(1) $\frac{9}{4}$

(2) $\frac{5}{3}$

(3) $\frac{4}{3}$

(4) $\frac{7}{4}$

Answer (1)

Sol. $P(A \cap B) = 0.1$, $P(A \mid B)$ and $P(B \mid A)$ are the roots of the equation $12x^2 - 7x + 1 = 0$

$$\Rightarrow P(A \mid B)\ P(B \mid A) = \frac{1}{12}$$

$$\Rightarrow \frac{P(A \cap B)}{P(B)} \times \frac{P(A \cap B)}{P(A)} = \frac{1}{12}$$

$$\Rightarrow P(A)\ P(B) = 12(0.1)^2$$

$$= 0.12$$

Also, $P(A \mid B) + P(B \mid A) = \frac{7}{12}$

$$\Rightarrow P(A \cap B)\left(\frac{1}{P(B)} + \frac{1}{P(A)}\right) = \frac{7}{12}$$

$$\Rightarrow P(A) + P(B) = \frac{7}{12} \times \frac{0.12}{0.1}$$

$$\Rightarrow P(A) + P(B) = 0.7$$

$$\frac{P(\bar{A} \cup \bar{B})}{P(\bar{A} \cap \bar{B})} = \frac{P(\overline{A \cap B})}{P(\overline{AUB})}$$

$$= \frac{1 - P(A \cap B)}{1 - P(A \cup B)}$$

$$= \frac{1 - 0.1}{1 - (0.7 - 0.1)} = \frac{0.9}{0.4} = \frac{9}{4}$$

16. Let $E : \frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$, $a > b$ and $H : \frac{x^2}{A^2} - \frac{y^2}{B^2} = 1$. Let the distance between the foci of E and the foci of H be $2\sqrt{3}$. If $a - A = 2$, and the ratio of the eccentricities of E and H is $\frac{1}{3}$, then the sum of the lengths of their latus rectums is equal to :

(1) 8

(2) 7

(3) 10

(4) 9

Answer (1)

Sol. $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ foci are $(ae, 0)$ and $(-ae, 0)$

$\frac{x^2}{A^2} + \frac{y^2}{B^2} = 1$ foci are $(Ae', 0)$ and $(-Ae', 0)$

$$\Rightarrow 2ae = 2\sqrt{3} \Rightarrow ae = \sqrt{3}$$

$$\Rightarrow 2Ae' = 2\sqrt{3} \Rightarrow Ae' = \sqrt{3}$$

$$\Rightarrow ae = Ae' \Rightarrow \frac{e}{e'} = \frac{A}{a}$$

$$\Rightarrow \frac{1}{3} = \frac{A}{a},\ a = 3A$$

Now, $a - A = 2 \Rightarrow a - \frac{a}{3} - 2 = 2$

$$\Rightarrow a = 3 \text{ and } A = 1$$

$$Ae = \sqrt{3} \Rightarrow e = \frac{1}{\sqrt{3}},\ e' = \sqrt{3}$$

$b^2 = a^2(1 - e^2)$

$b^2 = 6$

and $B^2 = A^2\left((e')^2 - 1\right) = 2$

$$\Rightarrow B^2 = 2$$

sum of L. R $= \frac{2b^2}{a} + \frac{2B^2}{A} = 8$



17. The sum of all values of $\theta \in [0, 2\pi]$ satisfying $2\sin^2\theta = \cos2\theta$ and $2\cos^2\theta = 3\sin\theta$ is

(1) π (2) $\frac{5\pi}{6}$

(3) 4π (4) $\frac{\pi}{2}$

Answer (1)

Sol. $2\sin^2\theta = \cos2\theta$ ...(i)

$2\sin^2\theta = 1 - 2\sin^2\theta$

$4\sin^2\theta = 1$

$\sin\theta = \pm\frac{1}{2}$

$2\cos^2\theta = 3\sin\theta$ ...(ii)

$2(1 - \sin^2\theta) = 3\sin\theta$

$2\sin^2\theta + 3\sin\theta - 2 = 0$

$\sin\theta = \frac{1}{2} \Rightarrow \theta = \frac{\pi}{6}, \frac{5\pi}{6}$

$\sin\theta = -2$ (Not possible)

$\because \ \theta = \frac{\pi}{6}, \frac{5\pi}{6}$

18. Let α_θ and β_θ be the distinct roots of $2x^2 + (\cos\theta)x - 1 = 0$, $\theta \in (0, 2\pi)$. If m and M are the minimum and the maximum values of $\alpha_\theta^4 + \beta_\theta^4$, then $16(M + m)$ equals:

(1) 17 (2) 24

(3) 27 (4) 25

Answer (4)

Sol. $2x^2 + (\cos\theta)x - 1 = 0$

$\alpha_\theta + \beta_\theta = \frac{-\cos\theta}{2}$

$\alpha_\theta \cdot \beta_\theta = \frac{-1}{2}$

$\alpha_\theta^2 + \beta_\theta^2 = (\alpha_\theta + \beta_\theta)^2 - 2\alpha_\theta\beta_\theta \frac{\cos^2\theta}{4} + 1$

$\alpha_\theta^4 + \beta_\theta^4 = \left(a_\theta^2 + \beta_\theta^2\right)^2 - 2\alpha_\theta^2\beta_\theta^2 = \left(\frac{\cos^2\theta}{4} + 1\right)^2 - \frac{2}{4}$

$= \left(\frac{\cos^2\theta}{4} + 1\right)^2 - \frac{1}{2}$

Maximum when $\cos\theta = 1$

$M = \left(\frac{1}{4} + 1\right)^2 - \frac{1}{2}$

$M = \frac{17}{16}$

Minimum when $\cos\theta = 0$

$m = 1 - \frac{1}{2} = \frac{1}{2}$

$16(M + m) = 16\left(\frac{17}{16} + \frac{1}{2}\right) = 25$

19. If $x = f(y)$ is the solution of the differential equation $\left(1 + y^2\right) + \left(x - 2e^{\tan^{-1}y}\right)\frac{dy}{dx} = 0, y \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$ with $f(0) = 1$, then $f\left(\frac{1}{\sqrt{3}}\right)$ is equal to:

(1) $e^{\pi/12}$ (2) $e^{\pi/6}$

(3) $e^{\pi/3}$ (4) $e^{\pi/4}$

Answer (2)

Sol. $\left(1 + y^2\right)\frac{dx}{dy} + x = 2e^{\tan^{-1}y}$

$\Rightarrow \frac{dx}{dy} + \frac{1}{1+y^2}x = \frac{2e^{\tan^{-1}y}}{1+y^2}$

I.F $= e^{\int \frac{1}{1+y^2}dy} = e^{\tan^{-1}y}$

Solution of $DE \Rightarrow x.e^{\tan^{-1}y} = \int \frac{2\left(e^{\tan^{-1}y}\right)^2}{1+y^2}dy$



$\Rightarrow x.e^{\tan^{-1}y} = e^{2\tan^{-1}y} + c$

$f(0) = 1 \Rightarrow c = 0$

$\Rightarrow x = e^{\tan^{-1}y}$

$f\left(1/\sqrt{3}\right) = e^{\tan^{-1}\left(\frac{1}{\sqrt{3}}\right)} = e^{\pi/6}$

20. If $\int e^x \left(\frac{x\sin^{-1}x}{\sqrt{1-x^2}} + \frac{\sin^{-1}x}{(1-x^2)^{3/2}} + \frac{x}{1-x^2} \right) dx = g(x) + C$,

where C is the constant of integration, then $g\left(\frac{1}{2}\right)$ equals:

(1) $\frac{\pi}{4}\sqrt{\frac{e}{2}}$ (2) $\frac{\pi}{6}\sqrt{\frac{e}{2}}$

(3) $\frac{\pi}{4}\sqrt{\frac{e}{3}}$ (4) $\frac{\pi}{6}\sqrt{\frac{e}{3}}$

Answer (4)

Sol. $\frac{d}{dx}\left(\frac{x.\sin^{-1}x}{\sqrt{1-x^2}}\right) = \sin^{-1}x \cdot \left(\frac{1.\sqrt{1-x^2} + \frac{x\cdot 2x}{2\sqrt{1-x^2}}}{1-x^2} \right) + \frac{x}{\sqrt{1-x^2}} \cdot \frac{1}{\sqrt{1-x^2}}$

$= \frac{\sin^{-1}x}{(1-x^2)^{3/2}} + \frac{x}{1-x^2}$

Hence, $I = \int e^x (f(x) + f'(x))dx$

$= e^x.f(x) + C$

$I = e^x \cdot \frac{x.\sin^{-1}x}{\sqrt{1-x^2}} + C = g(x) + C$

$\Rightarrow g(x) = \frac{xe^x \sin^{-1}x}{\sqrt{1-x^2}}$ and $g\left(1/2\right) = \frac{\pi}{6}\sqrt{\frac{e}{3}}$

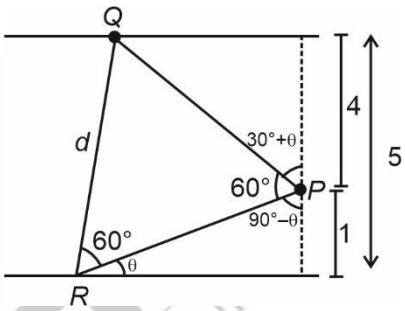
SECTION - B

Numerical Value Type Questions: This section contains 5 Numerical based questions. The answer to each question should be rounded-off to the nearest integer.

21. Let the distance between two parallel lines by 5 units and a point P lie between the lines at a unit distance from one of them. An equilateral triangle PQR is formed such that Q lies on one of the parallel lines, while R lies on the other. then $(QR)^2$ is equal to _____.

Answer (28)

Sol.



$PR = \frac{1}{\sin\theta}$ $PQ = \frac{4}{\cos(30° + \theta)}$

$\therefore$ PQR is equilateral.

$\Rightarrow \frac{1}{\sin\theta} = \frac{4}{\cos(30° + \theta)}$

$\frac{\sqrt{3}}{2}\cos\theta - \frac{1}{2}\sin\theta = 4\sin\theta$

$\Rightarrow \tan\theta = \frac{1}{3\sqrt{3}}$

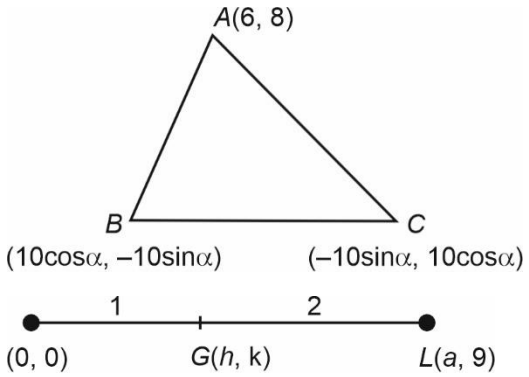
$QR^2 = d = \text{cosec}^2\theta = 28$

22. Let $A(6, 8)$ $B(10\cos\alpha, -10\sin\alpha)$ and $C(-10\sin\alpha, 10\cos\alpha)$, be the vertices of a triangle. If $L(a, 9)$ and $G(h, k)$ be its ortho center and centroid respectively, then $(5a - 3h + 6k + 100\sin 2\alpha)$ is equal to _____.

Answer (145)



Sol.



$\frac{a+0}{3} = h \Rightarrow a = 3h$

$\frac{9+0}{3} = k \Rightarrow k = 3$

$\because \ (h, k) = \left(\frac{6+10\cos\alpha-10\sin\alpha}{3}, \frac{8-10\sin\alpha-10\cos\alpha}{3}\right)$

$6 + 10\cos\alpha - 10\sin\alpha = 3h$

$10\cos\alpha - 10\sin\alpha = 3h - 6 \quad \ldots(1)$

$10(\cos\alpha - \sin\alpha) = 1 \quad \ldots(2)$

$\frac{8-10\sin\alpha+10\cos\alpha}{3} = k$

$\Rightarrow \ 100\sin2\alpha = 99$

$h = \frac{7}{3}$

$\Rightarrow \ a = 7$

Now, $5a - 3h + 6k + 100\sin2\alpha = 35 - 7 + 18 + 99$

$= 145$

23. Let $A = \{1, 2, 3\}$. The number of relations on A, containing (1, 2) and (2, 3), which are reflexive and transitive but not symmetric, is _____.

Answer (3)

Sol. R is reflexive $\Rightarrow R$ have (1, 1), (2, 2), (3, 3)

R is transitive

$\because \ (1, 2), (2, 3) \in R \quad \therefore \ (1, 3) \in R$

$\therefore \ R_1 = \{(1, 1), (2, 2), (3, 3), (1, 2), (2, 3), (1, 3)\}$

Clearly R_1 is reflexive and transitive but not symmetric.

Similarly,

$R_2 = \{(1, 1), (2, 2), (3, 3), (1, 2), (2, 3), (1, 3), (3, 2)\}$

$R_3 = \{(1, 1), (2, 2), (3, 3), (1, 2), (2, 3), (1, 3), (2, 1)\}$

Therefore, 3 relations are possible

24. If $\sum_{r=1}^{30} \frac{r^2\left({}^{30}C_r\right)^2}{{}^{30}C_{r-1}} = \alpha \times 2^{29}$, then α is equal to _____.

Answer (465)

Sol. $\sum_{r=1}^{30} \frac{r^2\left({}^{30}C_r\right)^2}{{}^{30}C_{r-1}}$

$\frac{r^2 \cdot 30!}{(30-r)!r!} \cdot \frac{30!}{(30-r)!r!} \times \frac{(r-1)!(31-r)!}{30!}$

$= \frac{30!(31-r)}{(r-1)!(30-r)!}$

$\Rightarrow \ \sum_{r=0}^{30} \frac{r^2({}^{30}C_r)^2}{{}^{30}C_{r-1}} = 30\sum_{r=0}^{30}(31-r)^{29}C_{r-1}$

$= 30\sum_{r=0}^{30}(31-r)^{29}C_{30-r}$

$= 30\sum_{r=0}^{30}[(30-r)+1]^{29}C_{30-r}$

$= 30\sum_{r=0}^{30}\frac{29}{(30-r)}(30-r)\cdot {}^{28}C_{29-r} + 30\sum_{r=0}^{30} {}^{29}C_{30-r}$

$= 30.29.2^{28} + 30.2^{29}$



$= 30.2^{28}(29 + 2) = (31 \times 15) \cdot 2^{29}$

$= 465.2^{29}$

$\therefore \quad \alpha = 465$

25. Let $y = f(x)$ be the solution of the differential equation $\frac{dy}{dx} + \frac{xy}{x^2-1} = \frac{x^6+4x}{\sqrt{1-x^2}}, -1 < x < 1$ such that $f(0) = 0$. If $6\int_{-1/2}^{1/2} f(x)dx = 2\pi - \alpha$, then α^2 is equal to_____.

Answer (27)

Sol. $\frac{dy}{dx} + \frac{xy}{x^2-1} = \frac{x^6+4x}{\sqrt{1-x^2}}, -1 < x < 1$

I.F $= e^{-\frac{1}{2}\int \frac{2x}{1-x^2}dx}$

I.F $= e^{\frac{1}{2}\ln(1-x^2)}$

I.F $= \sqrt{1-x^2}$

As $-1 < x < 1, \Rightarrow$ I.F $= \sqrt{1-x^2}$

$\therefore y\cdot\sqrt{1-x^2} = \int \frac{x^6+4x}{\sqrt{1-x^2}} \cdot \sqrt{1-x^2}dx$

$y\cdot\sqrt{1-x^2} = \int (x^6+4x)dx$

$y\cdot\sqrt{1-x^2} = \frac{x^7}{7} + 2x^2 + C$

Given $f(0) = 0$

$C = 0$

$\therefore y = \frac{x^7}{7\sqrt{1-x^2}} + \frac{2x^2}{\sqrt{1-x^2}}$

$\int_{-1/2}^{1/2} f(x)dx = \frac{1}{7}\int_{-1/2}^{1/2} \frac{x^7}{\sqrt{1-x^2}}dx + 2\int_{-1/2}^{1/2} \frac{x^2}{\sqrt{1-x^2}}dx$

$\int_{-1/2}^{1/2} f(x)dx = 0 + \int_{-1/2}^{1/2} \frac{2x^2}{\sqrt{1-x^2}}dx$

Put $x = \sin\theta$

$dx = \cos d\theta$

When $x = \frac{1}{2}, \theta = \frac{\pi}{6}$

$x = \frac{-1}{2}, \theta = \frac{-\pi}{6}$

$\int_{-\pi/6}^{\pi/6} \frac{2\sin^2\theta.\cos\theta d\theta}{\cos\theta}$

$= 2\int_{-\pi/6}^{\pi/6} \frac{(1-\cos 2\theta)}{2}d\theta$

$\int_{-1/2}^{1/2} f(x)dx = \frac{\pi}{3} - \frac{\sqrt{3}}{2}$

$6\int_{-1/2}^{1/2} f(x)dx = 2\pi - 3\sqrt{3}$

$\therefore \alpha = 3\sqrt{3}$

$\alpha^2 = 27$



PHYSICS

SECTION - A

Multiple Choice Questions: This section contains 20 multiple choice questions. Each question has 4 choices (1), (2), (3) and (4), out of which **ONLY ONE** is correct.

Choose the correct answer :

26. Given below are two statements. One is labelled as **Assertion (A)** and the other is labelled as **Reason (R)**.

Assertion (A) : A simple pendulum is taken to a planet of mass and radius, 4 times and 2 times, respectively, than the Earth. The time period of the pendulum remains same on earth and the planet.

Reason (R) : The mass of the pendulum remains unchanged at Earth and the other planet. In the light of the above statements, choose the **correct** answer from the options given below.

(1) **(A)** is false but **(R)** is true

(2) Both **(A)** and **(R)** are true and **(R)** is the correct explanation of **(A)**

(3) **(A)** is true but **(R)** is false

(4) Both **(A)** and **(R)** are true but **(R)** is **NOT** the correct explanation of **(A)**

Answer (4)

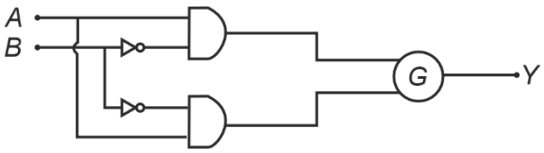
Sol. $T = 2\pi\sqrt{\frac{\ell}{g}}$

$$g = \frac{GM}{R^2}$$

$$g' = \frac{G\,4M}{(2R)^2} = g$$

Reason is true but not correct explanation.

27.



A	B	Y
0	0	1
0	1	0
1	0	0
1	1	1

To obtain the given truth table, following logic gate should be placed at *G*

(1) NOR Gate (2) NAND Gate

(3) OR Gate (4) AND Gate

Answer (Bonus)

Sol. As per the figure output is not matching with any of the options given.

Output for NOR gate is $\left(\overline{A}+B\right)$

Output for NAND gate is $\left(\overline{A}+B\right)$

Output for OR gate and AND gate is $A\overline{B}$

None of the option is matching

28. The torque due to the force $\left(2\hat{i}+\hat{j}+2\hat{k}\right)$ about the origin, acting on a particle whose position vector is $\left(\hat{i}+\hat{j}+\hat{k}\right)$, would be

(1) $\hat{i}+\hat{k}$ (2) $\hat{i}-\hat{j}+\hat{k}$

(3) $\hat{j}+\hat{k}$ (4) $\hat{i}-\hat{k}$

Answer (4)

Sol. $\tau = \vec{r}\times\vec{F}$

$$\tau = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & 1 & 1 \\ 2 & 1 & 2 \end{vmatrix}$$

$$\tau = \hat{i}(2-1) - j[0] + \hat{k}(1-2)$$

$$= \hat{i} - \hat{k}$$

